

Service Cutter: A Systematic Approach to Service Decomposition

Source: M. Gysel *et al.* "Service Cutter: A Systematic Approach to Service Decomposition," *Proceedings of the 5th European Conference on Service-Oriented and Cloud Computing (ESOCC 2016)*, 2016. [Online]. Available: http://link.springer.com/10.1007/978-3-319-44482-6_12

Paper Information: Definitions

Rating: ★★★★★

Own Keywords: Approaches, Background, Domain-Driven Design (DDD), Coupling Criteria

Reading Progress: First Pass

Introduction (What is the Paper about?)

- defining **Coupling Strategies in a catalog**
- present a tool called **Service Cutter** using that coupling strategies
- present useful definitions for the field

Methods

- **Service Cutter** extracts information from software engineering artifacts such as Domain Models and Use Cases
 - represent them as an uni-directed, weighted graph
 - find and score densely connected clusters
- **Validation Activities** including *Prototyping*, *Action Research (?)* and *Two Case Studies*
 - Performance Measurements
 - Feedback from members of the target audience in industry and academia

Scientific Benefit

- Collect **Coupling Strategies in a catalog** distilled from literature, industry and workshops
 - Catalog can serve to establish common terminology in discussions and documentation
- Introduced **Service Cutter** a knowledge management method and supporting tool framework assisting software architects when they making service design decisions
 - Can also suggest service decomposition candidates
- Most, but not all test scenarios resulted in appropriate service cuts
- Members from the target audience in industry and academia acknowledge that presented coupling criteria catalog and tool-supported service decomposition method have the potential to assist service architect's design decisions

Challenges / Future Work

Limitations

- not intended to fully automate Service Decomposition but rather support it
- collection might not yet be complete
 - (Own) Requirements on software might change in the future or might be weighted differently

Paper Outline and Read Progress

1. [X] Introduction
2. [X] Scopes the context and defines terminology
3. [X] Presentation of the Coupling Criteria Catalog
4. [] Definition of a novel service decomposition process and tool architecture
5. [] Presentation of an Implementation (*Service Cutter*) and validation
 - includes two case studies and performance measurements
6. [] Discussion of Strengths and Weaknesses of Service Cutter
7. [] Conclusion and Future Work

Own Comments

- (+) Key paper
- (+) Paper serves as foundation of the field of *Service Decomposition*
- (-) Validation mainly is based on peoples opinion (*Empirical*)
- (-) Coupling Criteria could be slightly different today (*Potentially outdated now*) including more or different criteria
 - Assume that they not change so fast, but this should be reevaluated